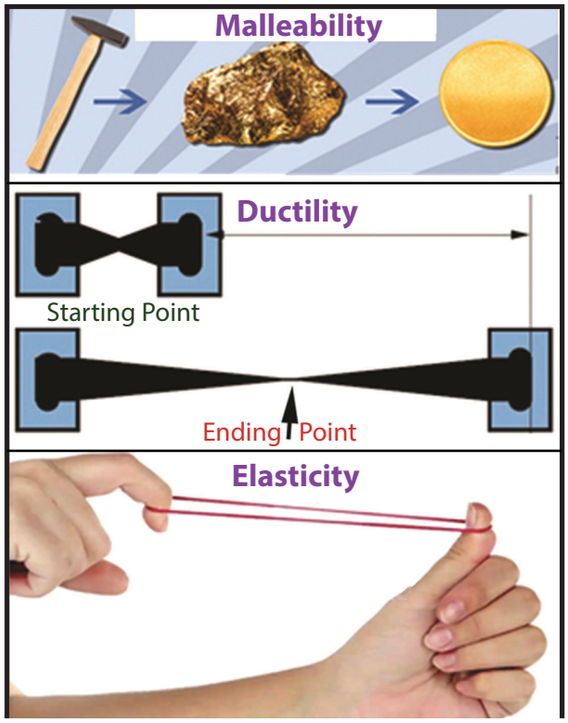
Malleability: Malleability is a property of solid material which Indicates that how easily a material gets changed in shape under compressive stress. Material can be formed into thin sheet by hammering (or) rolling. Copper, Aluminium and silver are malleable materials.

Ductility: Ductility is a property of solid materials which indicates that how easily a materials get changed in shape under tensile stress. Metals can be pulled into wire by using this property.

Elasticity: The ability of material to regain its original size and shape on removal of the applied forces is known as elasticity. Low carbon steels are having this elastic property. It is used to make spring.

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Malleability, Ductility and Elasticity

4.4 METALS

Metals are very useful in Industrial field. They are used to make tools because they can be strong and easy to share. Bridges, buildings or ships are constructed by using Iron and Steel. Most metals are heavy and they melt only when they are heated at very high temperature. Heat and electricity can easily pass through metals. A lump of metals can be beaten into a thin sheet or can be pulled into thin wires. Most of the metals are solid at room temperature except mercury. Mercury is liquid at room temperature.

Metals are classified as ferrous and non-ferrous metals.

Ferrous: Ferrous metals mostly contain Iron. They have small amounts of other metals or elements added, to give the required properties.

Types of ferrous metals

1. Pure Iron
2. Steel
3. Cast Iron

Pure Iron: Iron is a chemical element with symbols Fe and its atomic number is 26. It is an iron alloy with a very low carbon (0.1%) content. It is commercially known as pure Iron. It is relatively soft etc.



Pure Iron
Example Wrought Iron

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